

Air Quality Insights For Clarity Movement Co.

Causes, Effects, and Solutions



November 9th, 2025

*Understanding NO₂ and Ozone
relationships Across U.S Regions*

Business Problem

- Does high NO₂ affect ozone levels and air quality?
- Do pollution trends differ across regions?
- Where should interventions be prioritized?





Executive Summary

- Higher NO₂ → Lower O₃
- Regional differences do not influence trends
- Prioritize high NO₂ areas for action

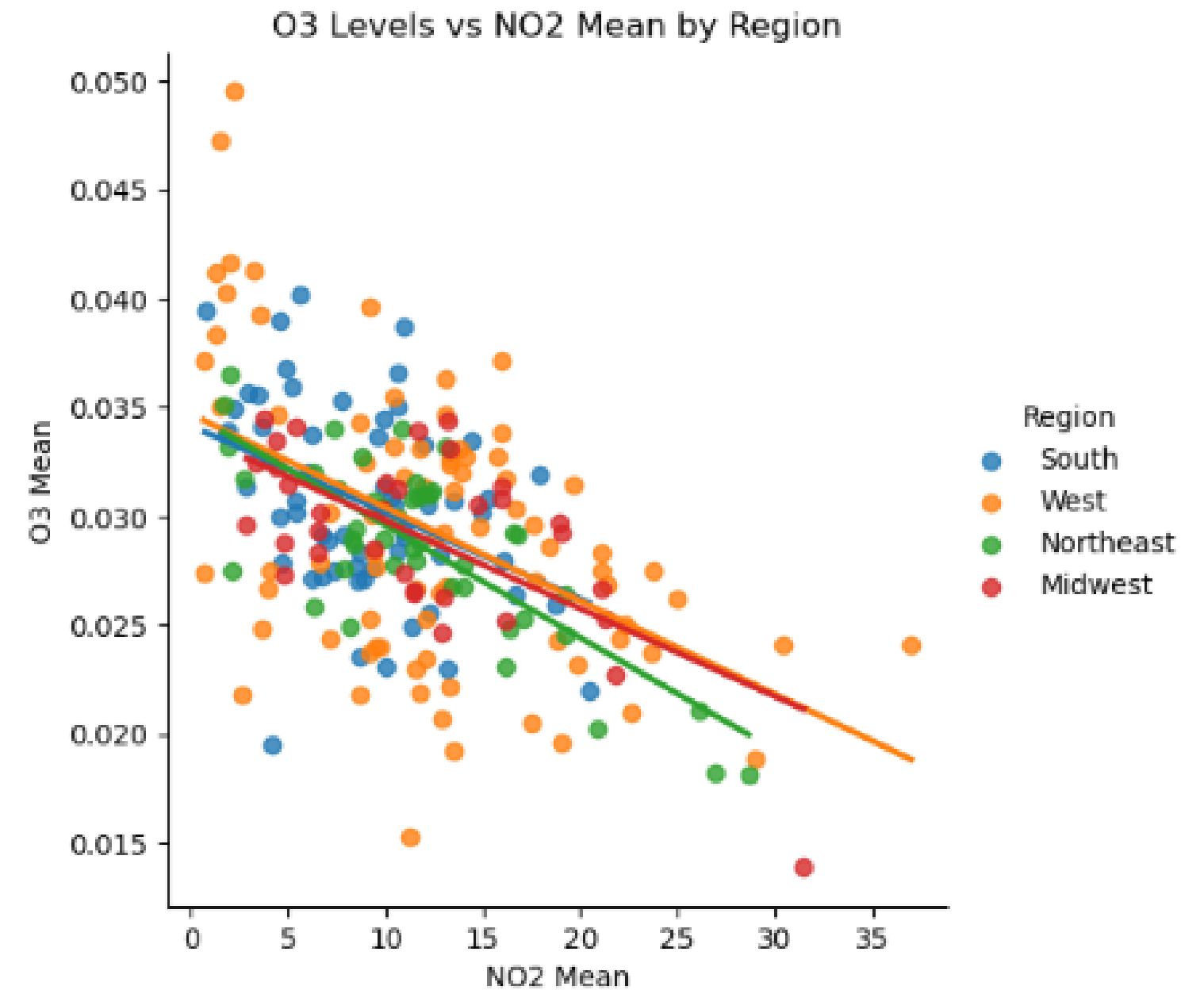
Data Overview

- EPA air quality data (2000–2022)
- Focus: O₃, CO, NO₂, SO₂ (mean values)
- Predicting ozone based on other pollutants and region



Pollution Trends by Region

- Scatterplot: NO2 vs O3, colored by region
- Clear negative correlation
- Regional differences minimal





Regression Results

- $\text{NO}_2 \rightarrow \text{O}_3$: strong negative relationship
- CO and SO_2 : weaker or insignificant
- Region: not significant
- Weaker $R^2 \rightarrow$ other factors may contribute

*Implement, monitor, and
optimize interventions*

- Launch targeted mitigation programs
- Monitor NO₂ and ozone trends
- Use insights for strategic planning

Next Steps

